

STANDARD OPERATING PROCEDURE

ELI LILLY AND COMPANY

North American Logistics Operations

US Drug Supply Chain Security Act (DSCSA)

Exceptions Management in North American

Logistics Operations (NALO)

USD-SOP-NALO US DSCSA Exceptions Management

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Section 1 — Document Administration

1.1 Purpose and Scope

For compliance and patient safety, this procedure details how NALO resolves DSCSA exceptions, including discrepancies between physical products and electronic transaction data.

1.2 Areas Involved

- North American Shared Service Center – Customer Service
- NALO Tech@Lilly
- NALO Operations
- Serialization SAP Tech@Lilly
- Global Serialization & Packaging (GSP)
- Lilly Value & Access
- NALO Quality Assurance
- Global Quality Manufacturing Logistics
- Consumer Product Quality Assurance (CPQA)

1.3 Approvals and Signatories

Approval of this procedure requires the signature of:

- Management, North American Shared Service Center – Customer Service
- Management, NALO Tech@Lilly
- Management, NALO Operations
- Management, Serialization SAP Tech@Lilly
- Management, Global Serialization & Packaging (GSP)
- Management, Lilly Value & Access
- Management, NALO Quality Assurance
- Management, Global Quality Manufacturing Logistics
- Management, Consumer Product Quality Assurance (CPQA)

1.4 Revision History

Revision	Reason for Change	Date
Version 1.0	New procedure	Oct, 2025
Version 2.0	• Service Now (SNOW) Reporting Requirements, including DSCSA AI SNOW Automation Tool Implementation • Overall triage flow context • Process for managing DSCSA Exceptions when no data is available • Process for managing swaps, including inventory verification for suspected swaps • Deviation Determination Criteria • Documentation and Closure Requirements	

1.5 Related Documents

- SRL-SOP-GSP Data Broker Error Management
- GQS105, Data Integrity and Good Documentation Practices
- USD-SOP-NASSC_CS Sales Order for Trade Product Entry-012
- GPR-501-5-109 Serialization TraceLink Partner Setup
- GPR-501-5-115 Decommissioning Serialization Material
- LQP-112-5, Requirements for Reporting Illegitimate Product and Falsified Medicines to Regulatory Authorities
- Product Quality Complaint (PQC) Complaint-handling SOP

Section 2 — Definitions, Acronyms, and Background

2.1 Regulatory Background — DSCSA and EDDS Requirements

The U.S. FDA Drug Supply Chain Security Act, often referred to as DSCSA, is a regulation designed to improve and ensure the safety of the U.S. pharmaceutical supply chain for interoperability.

The Enhanced Drug Distribution Security (EDDS) requirements of DSCSA require tracing of products at the unit level through the entire supply chain, leveraging serialization. Once EDDS is in effect, the Transaction Information (TI) should contain serial numbers that must match the physical product shipped. NALO has processes in place to efficiently manage exceptions (shipping errors, data issues) promptly to:

- Successfully investigate and resolve all issues
- Avoid wholesaler quarantining the product and pause further distribution of product
- Avoid shortage in the market
- Avoid the risk of wholesaler returning the product to Lilly if the data is incorrect or not provided

2.2 Definitions and Acronyms

Term	Definition
AS2	Applicability Statement 2 – Business-to-Business secure messaging protocol used to transmit Electronic Data Interchange (EDI) documents from one organization to another
ASN	Advanced Shipping Notice – submission of batch level details to the recipient of an order from the shipping location
ATTP	Advanced Track & Trace for Pharmaceuticals. Source of all serial numbers and ranges. It distributes numbers to packaging lines, palletizers, and the warehouse.
CARTS	Complaints and Remarks to Suppliers (and Service Providers). A formal quality communication tool used to notify suppliers where there is a confirmed issue, or potential issue.
DSCSA	Drug Supply Chain Security Act
EPCIS	Electronic Product Code Information Services; Industry standard for exchange of serialized data

GLN	Global Location Number – standardized number assigned to identify a company or a specific company location
GS1	International non-profit standards organization
GSP	Global Serialization Program
GTIN	Global Trade Identification Number
ICI	Intracompany Issue. Pre-deviation step to enable structured communication between sites. Help to assess if issues require a deviation.
LedgerDomain	Third Party owner of software used for receiving serial number verification requests
Lilly Sales Order	SAP sales order that includes sold-to, ship-to, products, and quantities
LVA	Lilly Value & Access
NALO	North American Logistics Operations
NTM	Notice to Management. Process for escalating issues.
Product Complaint	Any communication alleging a deficiency in the identity, quality, durability, reliability, safety, effectiveness, labeling, packaging, or performance of a Lilly product after it has been released for distribution or use.
QMS	Quality Management System. Integrated, global system of processes, standards, tools and governance.
REQ	Request – The overall request when you submit a Service Now request
RITM	Requested Item – The specific thing you're asking when you submit a Service Now request
SAP	Global Enterprise Resource Planning system
Serial Number	Unique number for each GTIN that identifies a saleable unit
SNOW	Service Now – Lilly enterprise platform used to submit, track, manage, and resolve work requests and issues
SGTIN	Combination of GTIN and Serial Number used to search in SAP to confirm a valid serial number. Syntax: 01[GTIN]21[Serial Number]
TLAC	The Lilly Answers Center
TraceLink	Third Party Cloud that serves as the Data Broker platform to manage all external partner data exchange for serialized product distribution
WMS	Warehouse Management System

2.3 Key Systems Overview — SAP and TraceLink Purpose

To achieve and maintain compliance with the product tracing requirements, Lilly has chosen third party software vendor, TraceLink, to provide the Lilly DSCSA Solution. The TraceLink Life Sciences Cloud is designed to fulfill US DSCSA requirements for the generation of T3s – combined Transaction Information, Transaction History, and Transaction Statement documents and their electronic storage (Section 582(b)(1)(C) of the FD&C Act). The solution conforms to current industry standards for lot-based tracking and utilizes the Healthcare Distribution Alliance (HDA) extended EDI 856 ASN for exchanging compliance information.

Using a combination of data entered by Lilly administrators and Lilly Master Data defined in the system during onboarding, the TraceLink Life Sciences Cloud generates complete, compliant transactions. The Lilly SAP master data system configuration systematically triggers interfacing to TraceLink of the data

elements supporting product first time shipment into the commercial supply chain. TraceLink will then map the data to the EDI ASN format and EPCIS format to be sent to wholesale distribution per DSCSA transaction tracing requirements.

Section 3 — Roles and Responsibilities

The DSCSA Triage Team serves as the first-line, cross-functional function that receives, assesses, coordinates, and drives resolution of DSCSA exceptions and verification requests to restore product-data alignment and maintain compliance and audit readiness. Detailed below are roles and responsibilities:

Role / Team	Key Responsibilities
Exceptions Resolution Coordinator – Intake	End-to-end orchestration of an exception once triaged, progress tracking across systems, drives closure communication with wholesalers, ensures documentation completeness
SAP ATTP Tech@Lilly	Investigates and corrects serialization events and aggregation issues in ATTP, fixing missing or mismatched serial data, retriggering EPCIS without reversing PGI to restore product data alignment
Data Broker Tech@Lilly	Investigates and resolves TraceLink-related issues, validating EPCIS/ASN transmissions, troubleshooting AS2/MDN messaging failures, confirming serialized data delivery and partner connectivity
NALO Customer Service	Enables commercial and financial resolution of DSCSA exceptions, manages credits, returns, reverse-logistics outcomes when compliance issues cannot be resolved through data correction alone
NALO Operations and 3PL Operations	Validates physical warehouse activities (pick, pack, ship, swaps, relabeling, inventory movements), confirms whether DSCSA exceptions are driven by operational handling versus data or system issues, helps DSCSA triage team to perform inventory count, logs deviation anytime a procedure is not followed that leads to DSCSA exception
NALO Quality Assurance	Provides DSCSA triage with compliance oversight, ensures proper deviation determination, documentation integrity, audit readiness, supports regulatory responses and quality governance
LVA Channel Operations and Trade Accounts	Manages trade and customer-facing implications of DSCSA exceptions, ensures compliant resolutions are communicated and executed in a way that aligns with wholesaler relationships and commercial policy
Global Serialization Program (GSP)	Provides serialization subject matter expertise, helps interpret serialization and aggregation behavior, assesses commissioning, decommissioning, rework scenarios, evaluates whether an exception is driven by serialization logic vs. operational execution
Global Quality – DSCSA	Accountable for overall compliance with DSCSA. Ensuring complete accuracy between physical and systematic representation of inventory and serial numbers. Accountable for quality control process in terms of recording the quality records to ensure that future exceptions can be minimized
Consumer Product Quality Assurance - CPQA	For any product return due to DSCSA exception, CPQA helps facilitate the return process from the downstream customer.

Section 4 — Exception Intake and Communication Standards

4.1 Single Intake Channel — Required Email Address and Routing Rules

All trade partner exceptions are submitted to Lilly_Data_Broker_DSCSA@lists.lilly.com. It is critical that this singular funnel is used to ensure awareness of issues and accountability of resolution with speed and priority. Any requests received internally through other pathways must immediately be redirected to this inbox. Work initiated outside of this directory should be redirected back to the Triage Team.

4.2 Exception Categories Defined

- Data Transmission Errors – Missing or incorrect transaction data (e.g., EPCIS files, ASN mismatches). Most common type, often requiring re-sending corrected data.
- Product/No Data – Physical product is received, but the corresponding serialized data is missing or delayed. This can trigger product holds until data is reconciled.
- Data/No Product – Serialized data is received, but the physical product is not. This may indicate a shipping or inventory error.
- Product/Data Mismatch – The serialized data does not match the physical product (e.g., wrong GTIN, serial number, or lot). This can lead to quarantines or returns.
- Packaging/Labeling Issues – Damaged or unreadable barcodes, incorrect labeling, or missing identifiers that prevent proper verification.

Section 5 — Overall Triage Flow and Process

5.1 Overall Triage Flow and Initial Evaluation

This section provides the high-level, end-to-end process overview for DSCSA exception management as performed by the DSCSA Triage Team within NALO. It serves as the master process map that governs the lifecycle of every DSCSA exception—from initial receipt through corrective action, preventive action, and incident closure—and establishes how the detailed sub-procedures documented elsewhere in this SOP are sequenced and interconnected.

All DSCSA exceptions follow this single process flow—see **Attachment A: DSCSA Exception Management Process High Level Overview**.

The following table maps each step in the high-level process flow to the applicable SOP section, enabling the reader to navigate from the overview to the detailed procedure.

Step	Phase	Activity
1	Intake	Receive incoming email for exception from customer via Lilly_Data_Broker_DSCSA@lists.lilly.com
2	Intake	DSCSA AI ServiceNow Automation Tool automatically processes and creates the appropriate records (INC, INCTSK) in ServiceNow for every new exception
3	Triage	Decision: Are we able to provide data?
4	Returns (NO)	Log a complaint with TLAC to initiate the return process with CPQA
5	Corrective Actions (YES)	Start the investigation process to find the root cause

6	Corrective Actions	Once root cause is identified, confirm the steps to resolve the DSCSA data exception
7	Corrective Actions	Take the necessary steps in SAP ATTP / S4 / TraceLink to revise the data and send EPCIS data to the customer
8	Corrective Actions	Retrieve confirmation from the customer on receipt and acceptance of data
9	Preventive Actions	Decision: Was the root cause due to a process issue or system issue?
10a	Preventive (System)	Enter the needed Incident / RITM in SNOW, if applicable
10b	Preventive (Process)	Decision: Was this issue caused by Internal Site or External Site?
11a	Preventive (Internal)	Create a deviation / ICI to cover the exception (or add it to an existing applicable deviation)
11b	Preventive (External)	Create a CARTS for the supplier (or add it to an existing applicable CARTS)
12	Closure	Close the incident with correct resolution code, likely cause, and deviation / ICI / CARTS reference number (if applicable)

5.2 Use of ServiceNow to Manage the Process

All emails received at Lilly_Data_Broker_DSCSA@lists.lilly.com are required to be formally tracked. Upon receipt, the Lilly team documents each exception by creating an Incident in ServiceNow. It's critical that all Incidents accurately reflect the exception details as submitted by downstream partners. This helps to ensure appropriate response in a timely manner.

Section 6 — Exception Triage Workflow

6.1 Exception Types

Type 1: EPCIS/ASN Data Missing for Entire Shipment (Truck, No Data)

- Check TraceLink and ensure it's delivered; if not, investigate further
- ZSHP is retriggered at times to resend the data
- TraceLink is used to send ASN. If for any reason ASN is not sent, please follow up and investigate on TraceLink / SAP side

Type 2: Product Received, No Serialized Data (Product/No Data)

- These situations may occur due to discrepancies such as the application of incorrect packaging labels, physical swaps between deliveries or overage. Physical swaps can take place at either the carton/case level or the pallet level.
- In case of overage, NALO Trade Claims will receive the relevant claims. The DSCSA team will provide accurate data corresponding to the overage claim, and NALO Trade Claims will process a rush order without a pick delivery.
- For shipments missing the appropriate SGTIN label at the pallet or case label, customers should be requested to perform scanning at a lower packaging level.

Inability to Provide Serialization Data

In situations where we cannot provide serialization data, we process a return for the product. These are categorized into two buckets—product complaints and saleable returns linked to a different customer.

Product Complaint — See Attachment B

- DSCSA Triage collects relevant data and assesses the eligibility of product returns, such as when products cannot be scanned by customers or when decommissioned products are received.
- A Product Complaint (PC) must be initiated by the DSCSA team within 24 hours with The Lilly Answers Center (TLAC) to initiate a product return from the wholesaler.
- TLAC submits a Product Complaint (PC) for CPQA to process.
- CPQA updates the customer record and coordinates the return process.
- NALO Trade Claims administers customer credit once the return has been received and verified.
- The DSCSA Team is responsible for decommissioning the Serial Number, if applicable.

Saleable Returns Linked to a Different Wholesaler — See Attachment C

- Conduct DSCSA Triage shipment verification to determine whether the product was shipped to an alternate customer.
- Obtain delivery information from the requesting customer.
- If the customer does not provide valid delivery information, inform them that Lilly is unable to supply DSCSA transaction data for this product.

Type 3: Data Received, No Physical Product (Data/No Product) – Shortage

- The NALO Trade Claims team will provide a list of serial numbers identified as short.
- The NALO Trade Claims team will collaborate with NALO Operations and 3PL to determine and confirm the shortage.
- Upon confirmation, the Data Broker Team will initiate an incident to reactivate those serial numbers in both TL and SAP ATTP systems.

Type 4: Master Data Exceptions

In situations where EPCIS is not delivered because of GLN or NDC format issues, master data issues are resolved by the SAP / TraceLink teams via the Incident process.

Section 7 — Quality Oversight and Deviation Determination

A deviation is defined as a departure from approved processes, procedures, instructions, specifications, or regulatory requirements. This definition is inclusive of an unexpected occurrence where there is no procedural response, that is, a situation arises for which no existing SOP step, work instruction, or decision branch provides direction.

Each functional area is responsible for investigating and determining if deviation is needed and creating deviation record, if appropriate. In situations where there isn't clear functional ownership, the exceptions resolution coordinator will execute the deviation record.

Not every exception is a deviation. Routine exceptions that are resolved through standard triage (e.g., a successful data re-send, a trading partner configuration correction, or a confirmed non-Lilly issue) do not require deviation initiation.

USD-SOP-NALO US DSCSA Exceptions Management

The following matrix maps each exception type to the conditions under which a deviation should be initiated, the expected root cause category, and the resulting quality record type.

Exception Type	Deviation Trigger Condition	Root Cause Category	Quality Record Type
Product / No Data	EPCIS confirms product shipped but transaction data not transmitted; cannot be resolved by re-send	SAP Issue / TL Master Data	Deviation → CAPA if systemic
Product / Data Mismatch	Serial number, GTIN, lot, or expiry in transaction data does not match physical product; discrepancy confirmed after investigation	SAP Issue / Serialization Error	Deviation → CAPA if systemic
Data Transmission	Repeated transmission failures to the same trading partner after standard troubleshooting; or systemic EPCIS / AS2 connectivity issue	TL Master Data / Trading Partner	Deviation (Lilly-side) or Trading Partner notification
Label / Packaging	Physical label or packaging anomaly that prevents serialization verification or raises suspect product concern	Manufacturing / Packaging	Deviation → escalate per LQP-112-5 if suspect
Data / No Product	Transaction data sent but trading partner confirms product not received; confirmed after shipment investigation	Logistics / Warehouse Error	Deviation → CAPA if systemic; possible loss/theft investigation
Swap / Incorrect Shipment	Investigation confirms product was shipped to wrong customer or serial numbers were cross assigned between shipments	SAP Issue / Warehouse Error	Deviation → CAPA; return initiated via Trade Claims
No Procedural Response	An unexpected occurrence for which no existing SOP step, work instruction, or decision branch applies	To be determined during investigation	Deviation (mandatory); escalate to NALO Quality for assessment

Section 8 — Escalation Pathway

This section defines the tiered escalation framework for DSCSA exceptions, from initial resolution through FDA reporting. It establishes the criteria for advancing an exception through each escalation tier, the activities and documentation required at each level, and the standard exception type definitions used across all DSCSA exception management processes.

The escalation pathway is progressive: not every exception advances beyond Tier 0. The framework ensures that routine exceptions are resolved efficiently while significant issues receive the quality, stakeholder, and regulatory attention they require.

The escalation framework consists of five tiers (Tier 0 through Tier 4), each representing an increasing level of organizational response, quality system engagement, and regulatory exposure. Exceptions enter at the tier appropriate to their severity and may advance through higher tiers as investigation findings

warrant.

Tier	Escalation Level	Trigger Criteria	Key Activities	Documentation
Tier 0	No Response Needed	Timely issue resolution within DSCSA timing requirements. No further response is required.	Standard triage and resolution per exception type procedures	
Tier 1	ICI / CART / RITM or SNOW Response	Exception requires formal tracking but can be resolved without quality system escalation.	Create and manage ICI, CART, RITM, or SNOW record; coordinate with responsible party for resolution	Issue tracking in TraceLink required. ICI / CART / QMS / SNOW / RITM numbers tracked in SNOW record
Tier 2	Deviation Initiation	Exception meets deviation threshold (departure from approved process, GxP, or regulatory requirement).	Initiate deviation per NALO Quality Assess / Review / Investigate / Approve workflow	Deviation record in quality system and reference number included in SNOW record
Tier 3	Escalation Response – Stakeholder Meeting	Deviation investigation reveals potential impact to product quality, patient safety, or supply chain integrity that requires cross-functional decision-making.	Convene escalation stakeholder meeting; assess scope of impact; determine if FDA engagement is required	
Tier 4	3911 FDA Reporting / Consult	Confirmed impact to product integrity or confirmed illegitimate product requiring FDA notification.	Initiate 3911 FDA reporting process and/or FDA consult; execute NTM process if applicable	Documentation in Qdocs / FDA website

Section 9 — Incident Closure and Post-Resolution Requirements

Once a DSCSA exception has been resolved, the following activities must be completed to ensure proper documentation, customer confirmation, and regulatory compliance.

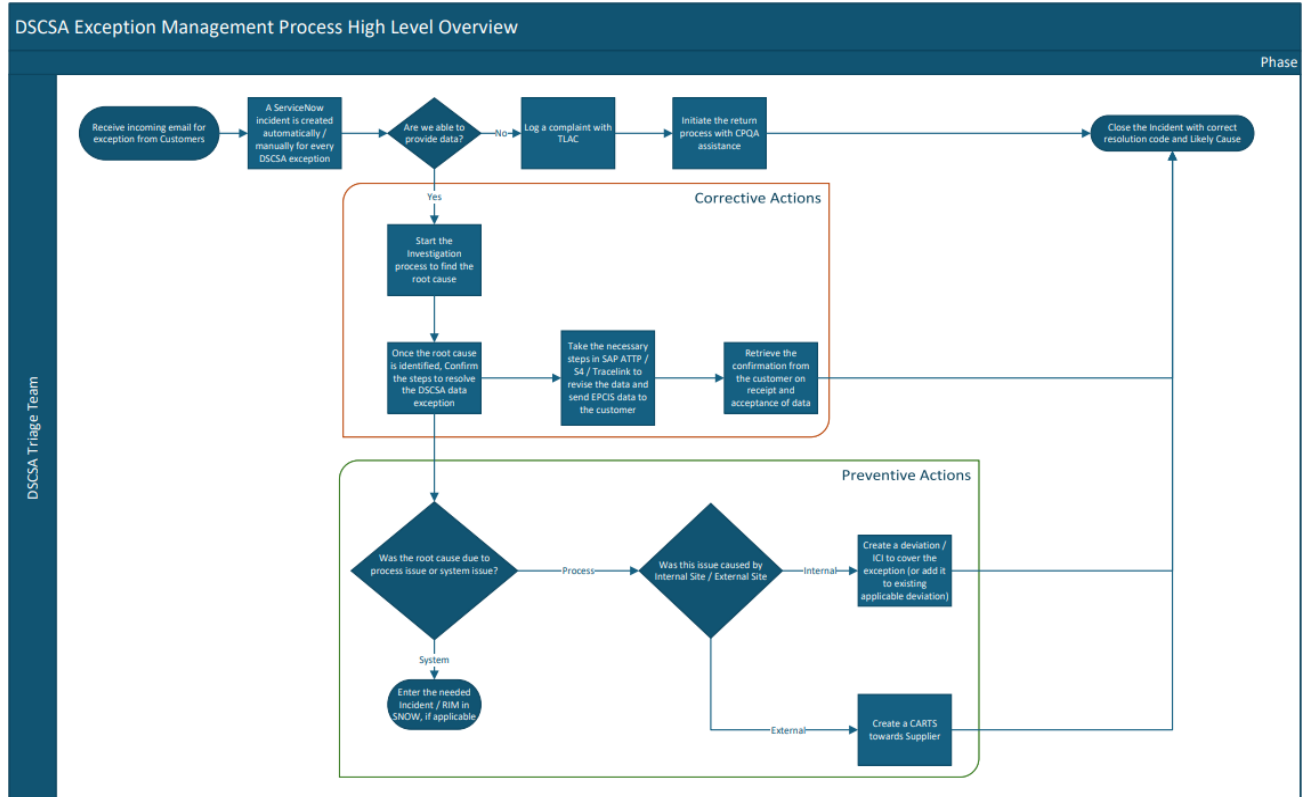
- Ensure the incident is closed only after the exception has been fully resolved.
- Confirm and record that the customer is satisfied with the resolution.
- Initiate a deviation or CARTS record if applicable and include reference number in SNOW record.
- Begin the return process in cases where DSCSA data cannot be furnished and include the Product Complaint number in SNOW record.

Section 10 — Serial Number Decommissioning

- The DSCSA Triage Team shall initiate a Request Item (RITM) to the NALO Quality Team to decommission the applicable serial number(s) in the SAP system. The RITM shall be assigned to assignment group APPS-SUPP-US-NALO-DSCSA-EXCEPTIONS.
 - RITM details should correctly reference the original ICI/Deviation/PC number to ensure linkage between the serial numbers being decommissioned and the exception.
 - Note: The serial numbers must bin SGTIN form to execute the decommissioning process in SAP.
- An RITM Task shall be created concurrently and assigned to the TraceLink Data Broker Team (assignment group DATABROKER-SERIALIZATION-GLB). The Data Broker Team shall perform serial number status confirmation in TraceLink only upon successful completion of the SAP decommissioning steps.
- These SAP steps are guided by the Global Decommissioning Serialization Material SOP
 - GPR-501-5-115

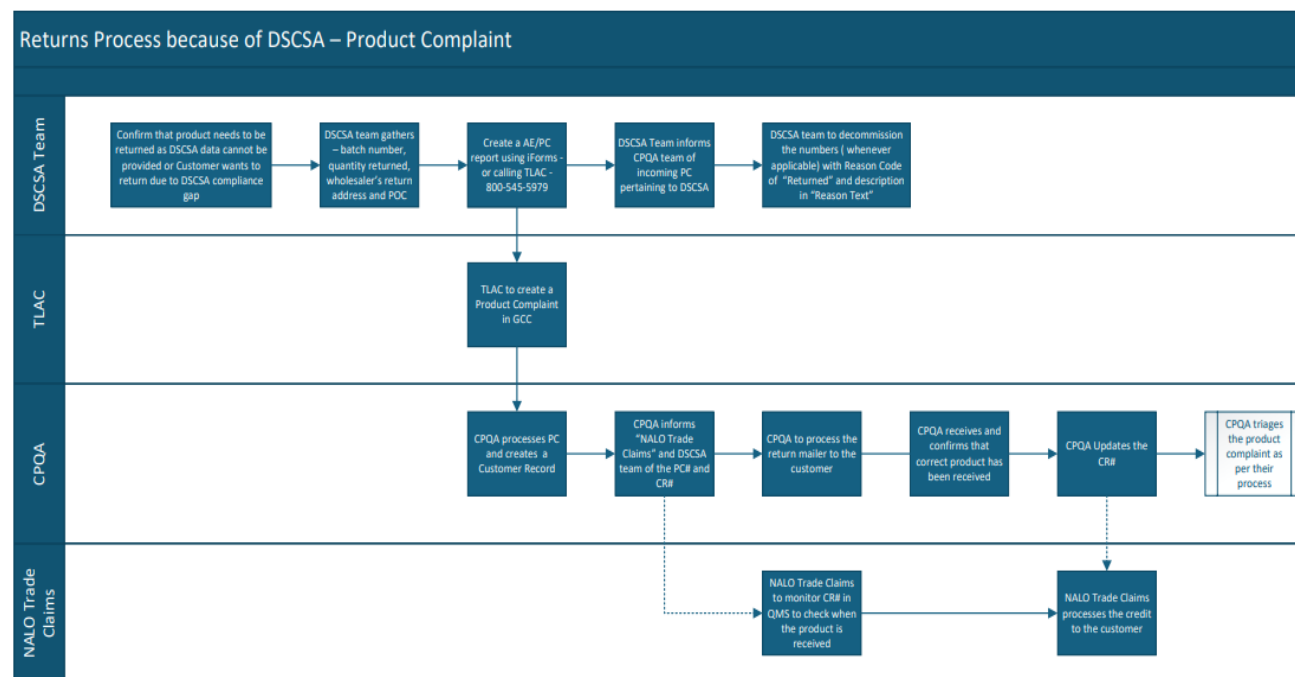
Attachments

Attachment A — DSCSA Exception Management Process High Level Overview



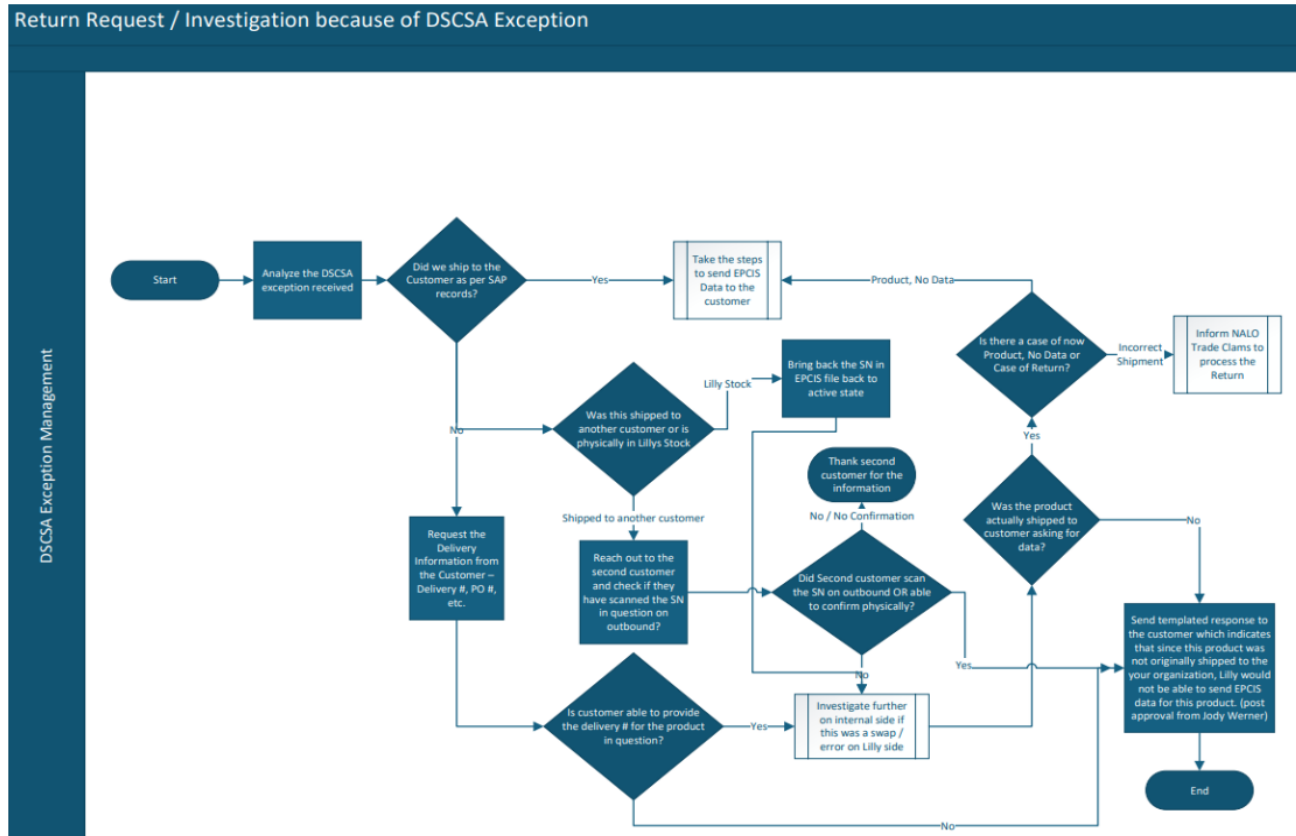
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Attachment B — Product Complaint Product Return



USD-SOP-NALO US DSCSA Exceptions Management

Attachment C — Saleable Return Product Return



Document Approval Signatures

Approved Date: 29 Jul 2026

Approval Verdict: Approve	Scott Dalton (RC94736@lilly.com) Approver 28-Jul-2026 15:15:40 GMT+0000
Approval Verdict: Approve	Deborah Boggs (AA88886@lilly.com) Approver 28-Jul-2026 15:31:45 GMT+0000
Approval Verdict: Approve	Sabry Samuel (C039290@lilly.com) Quality Approver 28-Jul-2026 15:43:04 GMT+0000
Approval Verdict: Approve	Jamil Shabazz (U7X1229@lilly.com) Approver 28-Jul-2026 16:25:06 GMT+0000
Approval Verdict: Approve	Eric Thomas (L107260@lilly.com) Approver 28-Jul-2026 16:48:58 GMT+0000
Approval Verdict: Approve	Taylor Wong (C160157@lilly.com) Approver 28-Jul-2026 17:00:15 GMT+0000
Approval Verdict: Approve	Brandon Bowler (L004253@lilly.com) Approver 28-Jul-2026 17:30:10 GMT+0000
Approval Verdict: Approve	Sean Couch (L090553@lilly.com) Approver 28-Jul-2026 18:27:17 GMT+0000
Approval Verdict: Approve	Nerea Goicoechea (YS09540@lilly.com) Approver 29-Jul-2026 08:55:04 GMT+0000
Reviewer / Approver Additional Details	N/A

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